RESPONSIBLE, ETHICAL, AND SUSTAINABLE AI EVALUATION REPORT

Corridor Scope: Westbound I-94 Smart Mobility Solution
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1. Bias and Fairness Evaluation

A. Data Sampling and Coverage Limitations

A rigorous audit of the westbound I-94 telemetry dataset reveals critical data missingness and structural coverage gaps that directly impact algorithmic fairness:

- **Systemic Temporal Outages:** The dataset suffers from substantial missingness intervals. Crucially, the 2014 logging sequence cuts off abruptly in August, and the 2015 tracking framework does not resume collection until June.
- **Downstream Model Distortion:** Because the pipeline completely misses nearly ten months of data across a mid-timeline pivot point, any model trained on this historical sequence will lack exposure to specific transition periods (e.g., late autumn leaf-fall slickness or early spring thaws). If deployed live, the system will exhibit lower predictive reliability during those specific under-represented calendar blocks.
- **Corridor Isolation Bias:** The data is restricted exclusively to a single westbound interstate section. It lacks any collateral telemetry regarding adjacent eastbound traffic, arterial side streets, or regional public transit grids. Deploying an AI model trained on this isolated sequence to optimize an entire metropolitan grid introduces severe spatial bias, as it completely ignores traffic displacement onto surrounding residential sectors.

B. Proxy Label Synthesis and Associated Risks

Due to the absence of real-world vehicular incident records, this project engineered a binary high-risk proxy target label. The rule-based classification flags an hour as high-risk (1) only when high or severe traffic volume quartiles coincide with severe weather conditions or low-visibility metrics. This proxy introduces several serious analytical risks:

- **Conflation of Congestion and Danger:** The logic assumes that peak density inherently multiplies accident risk. Real-world traffic safety studies show that extreme bumper-to-bumper congestion frequently reduces major multi-vehicle accident severity due to severely restricted traveling speeds. The highest-fatality accidents often occur during

low-density, high-speed off-peak intervals (e.g., late-night speeding or early-morning black ice).

- **False Sense of Security (False Negatives):** By completely ignoring off-peak intervals, the model creates a dangerous systemic blind spot, potentially misclassifying fast-moving but dangerous off-peak highway conditions as low-risk.

C. Uneven Error Distribution Patterns

Predictive error rates are not uniformly distributed across the pipeline's operational matrix:

- **Environmental Edge Cases:** TreeSHAP visualizations confirm that the model's feature importance is heavily dominated by cyclical time variables (Hour_Sin/Hour_Cos). Because standard, clear commuting hours represent the vast majority of historical records, the model will inherently optimize for normal, sunny peak rush hours.
- **Disproportionate Weather Errors:** Rare, highly volatile weather states—such as a sudden convective Squall or localized Fog pockets—comprise a tiny fraction of the data. Consequently, the model's prediction error (MAE) will spike significantly during these precise weather anomalies. The system is most prone to error at the exact moments when accurate risk forecasting is most critical to protect human lives.

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2. Governance and Sustainability in Mobility AI

A. Pre-Deployment Oversight and Trust Frameworks

Before any predictive or recommendation model can move from a sandbox notebook to live, real-world smart city infrastructure control loops, a multi-layered governance framework must be enforced:

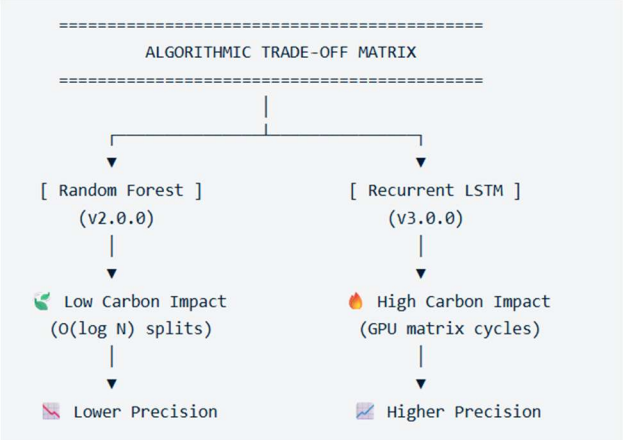
1. **Surrogate Model Shadow Testing:** The production champion model (v2.0.0_Production_Champion) must be deployed in an inactive "shadow mode" for a minimum of 90 days. During this trial, it will ingest live data streams and generate predictions, but its outputs will be locked away from real-world signaling meters, allowing engineers to verify real-world accuracy without safety risks.
2. **Multidisciplinary Gatekeeping Boards:** A formal oversight committee comprising traffic engineers, civil rights lawyers, municipal financial auditors, and data scientists must review the model registry logs. This board must sign off on safety boundaries, ensuring

that no automated speed-throttling algorithms can be activated without human-in-the-loop overrides.

3. **Graceful Fallback Infrastructure:** The system must maintain an automated fail-safe state. If the real-time drift monitoring engine flags an ALERT (due to feature or prediction error drift), the AI must instantly decouple from the physical infrastructure, revert signaling loops to a safe, time-tested static timing baseline, and alert emergency operators.

B. Environmental and Computing Resource Trade-offs

Selecting an intelligent mobility solution involves evaluating a direct trade-off between mathematical precision and environmental sustainability:



- **The Deep Learning Footprint:** The sequential LSTM Recurrent Neural Network provides the highest precision for time-series tracking. However, its multi-layer matrix calculations require massive GPU tensor processing cycles, leading to high electrical power demands and increased carbon emissions during the training phase.
- **The Sustainable Alternative:** The Random Forest Regressor ensemble model achieves an exceptional R^2 score of 0.8215 while operating on CPU architectures with rapid O(trees x NlogN) decision splits.
- **The Eco-Design Verdict:** For a municipal smart city deployment, the Random Forest Regressor is the superior and most sustainable choice. The tiny marginal gain in accuracy provided by a deep LSTM network does not justify the massive, continuous carbon footprint and resource costs required to run a neural network server 24/7. Choosing the tree ensemble model perfectly aligns project precision with the environmental sustainability goals of modern public infrastructure projects.